

Experiment 2:

Candidate-Elimination Algorithm Code:

```
import pandas as pd

# Read training data from CSV file
data = pd.read_csv("trainingdata.csv")

concepts = data.iloc[:, :-1].values
target = data.iloc[:, -1].values

# Initialize Specific Hypothesis
S = concepts[0].copy()

# Initialize General Hypothesis
G = [["?" for _ in range(len(S))] for _ in range(len(S))]

print("Initial Specific Hypothesis:")
print(S)

print("\nInitial General Hypothesis:")
for g in G:
    print(g)

# Candidate Elimination Algorithm
for i, h in enumerate(concepts):
    if target[i] == "Yes":
        for x in range(len(S)):
            if h[x] != S[x]:
                S[x] = "?"
                G[x][x] = "?"
    else:
        for x in range(len(S)):
            if h[x] != S[x]:
                G[x][x] = S[x]
            else:
                G[x][x] = "?"

print("\nFinal Specific Hypothesis:")
print(S)

print("\nFinal General Hypothesis:")
for g in G:
    if g != ["?" * len(S)]:
        print(g)
```

Output:

```
===== RESTART: E:/Machine Learning/candidate-elimination.py =====  
Initial Specific Hypothesis:  
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']  
  
Initial General Hypothesis:  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
['?', '?', '?', '?', '?', '?']  
  
Final Specific Hypothesis:  
['Sunny' 'Warm' '?' 'Strong' '?' '?']  
  
Final General Hypothesis:  
['Sunny', '?', '?', '?', '?', '?']  
['?', 'Warm', '?', '?', '?', '?']
```